

Conference on Nonlinear Systems and Dynamics

The background of the entire page is a dark, textured field filled with a complex network of thin, light-colored lines and small, semi-transparent circular nodes. These elements are interconnected, creating a web-like or molecular structure that suggests themes of systems, dynamics, or networks. Overlaid on this background is the text 'CNSD 2026' in a large, bold, cyan-colored font. The '6' at the end of the year is particularly large and stylized, with a slight shadow or glow effect.

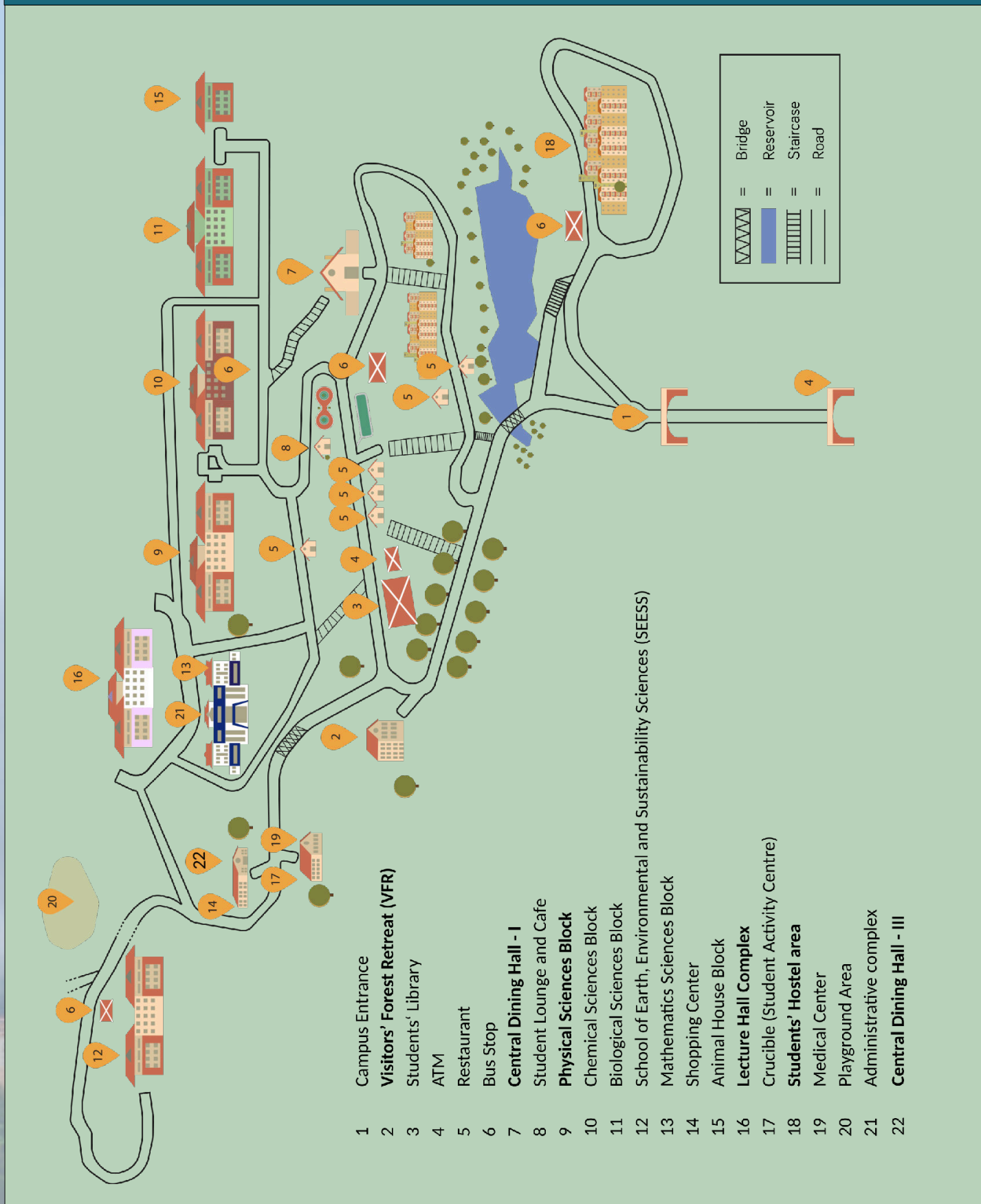
CNSD
2026

7th-10th July 2026, IISER Thiruvananthapuram, India

Invited Speakers

- Jürgen Kurths, PIK, Germany
- Stefano Boccaletti, CNR-ISC, Italy
- Eckehard Schöll, TU Berlin, Germany
- Rajarshi Roy, Univ. Maryland, USA
- Peter Ashwin, Univ. Exeter, UK
- Ulrike Feudel, Univ. Oldenburg, Germany
- Miguel AF Sanjuán, URJC, Spain
- Reik Donner, Magdeburg-Stendal UAS, Germany
- Boris Malomed, Tel Aviv Univ., Israel
- Sandip George, Univ. Aberdeen, UK
- Aasifa Rounak, Univ. College Dublin, Ireland
- Andrew Pickering, URJC, Spain
- Pilar Ruiz Gordoa, URJC, Spain
- Tomasz Kapitaniak, TUL, Poland
- Lakshmanan Muthusamy, BDU, Tamil Nadu
- Ram Ramaswamy, IISER Berhampur
- Soumitro Banerjee, IISER Kolkata
- Somdatta Sinha, IISER Mohali
- Avinash Khare, SPPU, Pune
- Sudeshna Sinha, IISER Mohali
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- K. Murali, Anna Univ., TN
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- R. Gopal, SASTRA Univ., TN
- Soumik Bandyopadhyay, IISER TVM
- Thounaojam Umeshkanta Singh, Bennett Univ.

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Parasite enrichment drives inequality before collapse in a model of evolving networks

Atiyab Zafar¹, Sanjay Jain^{1,2}

¹ Department of Physics and Astrophysics, University of Delhi, Delhi, India

² Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, New Mexico 87501, USA

Many complex systems, including stock markets and civilizations, can undergo collapse, and understanding why this happens is important for forecasting and response planning. Here, we study collapses in a mathematical model [1] of an evolving directed network with ($S \gg 1$) nodes. In the model, the eigenvector centrality (X_i) of node i is taken as the node's "strength." The network evolves one node at a time by replacing the weakest node (the node with the smallest (X_i)) with a new node that connects randomly to the existing network with connection probability (p). As the network evolves, the model shows spontaneous growth, stasis, and collapse in the number of nodes with non-zero strength, where a collapse corresponds to a sudden drop in this number. Previous studies have identified proximate causes of collapse in this model, [2, 3, 4] but the long-term dynamics that makes the network fragile and susceptible to collapse remains unclear. To probe this long-term approach to collapse, we compute the Gini coefficient, a standard measure of inequality, for the distribution of node strengths (X_i). We find that inequality rises on average as the network approaches collapse. We also find that the strength of peripheral parasitic nodes increases as collapse is approached; these nodes become richer at the expense of the core. Peripheral nodes are "parasitic" in the sense that they benefit from the core that sustains them but do not provide feedback that supports the core. This shift of strength away from the core makes the network more fragile and more prone to collapse. These lessons from the model and the above mentioned early warning signals of collapse may be relevant for other complex adaptive systems.

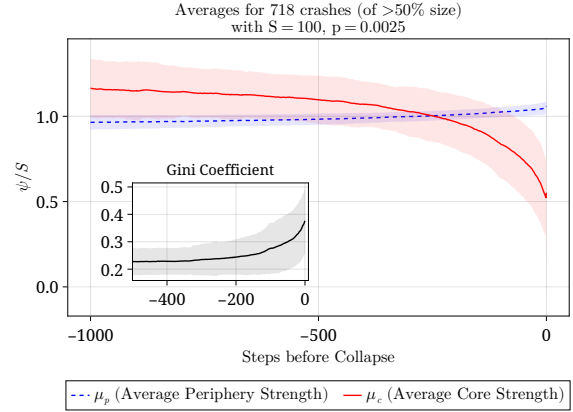


Figure 1: Periphery getting richer and rising inequality before collapse. The x -axis shows time in graph updates, with collapse at zero. Quantities are averaged over 718 runs ($S = 100$, $p = 0.0025$), each ending in a collapse. The red solid (blue dashed) curve shows the mean μ_c (μ_p), the average strength of core (periphery) nodes, across all runs. The blue dashed curve crossing and exceeding the red solid curve indicates parasites getting richer before collapse. The inset shows the Gini coefficient averaged over the same runs, representing rising inequality. Shaded bands (pink, blue, grey) denote one standard deviation of μ_c , μ_p , and the Gini coefficient respectively.

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IISER Thiruvananthapuram
Maruthamala P. O. Vithura,
Thiruvananthapuram -695551,
Kerala, India

